# SwiftLog Logistics: Executive Business Intelligence Report

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## 1. Executive summary

SwiftLog asked for a single view of its sales, deliveries, inventory, suppliers and returns. We cleaned eleven datasets, connected them in Power BI, built KPIs in DAX and produced a five-page interactive dashboard. The headline finding is a delivery problem: only **36.3%** of shipments arrive on time and the average delay is **3.6 days**. The problem is almost identical in every city and warehouse, which points to a company-wide process issue rather than one weak location. Delayed delivery is also the most common reason for returns. Delivered orders generated about **N7.11M** in revenue from **1,658** orders, with an average order value of about **N4,290**. Inventory is large but moves slowly, and 264 warehouse and product combinations are below the low-stock threshold.

## 2. Data preparation

All eleven files were imported, headers promoted, data types checked and text trimmed. Key steps: order dates converted to dates; a Date table added; Delivery Performance merged into Shipments; a Delay flag created (3 or more delay days); line revenue calculated as quantity multiplied by unit price; inventory summed per warehouse and product; and relationships built in a star-style model with Orders and Shipments at the centre. No broken keys or missing values were found.

## 3. KPI summary

| KPI | Value | Reading |
|---|---|---|
| Total revenue (delivered) | N7.11M | Revenue counts delivered orders only |
| Total orders / customers | 5,000 / 991 | Only 1,658 orders are delivered |
| Average order value | N4.29K | Delivered revenue / delivered orders |
| On-time delivery rate | 36.28% | Needs immediate attention |
| Average delay | 3.57 days | Proxy for delivery time |
| Return rate | 9.5% | Share of orders with a return |
| Inventory turnover | 0.03 | Units sold / stock held; very slow |
| Customer retention | 48.3% | Customers with 2+ delivered orders |

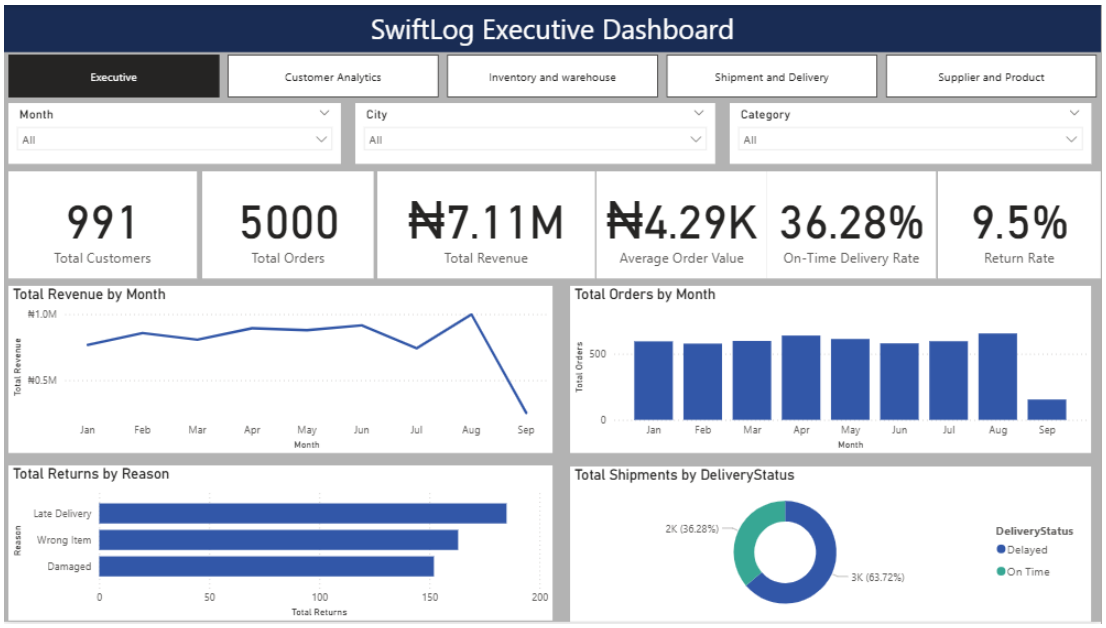



Figure 1: Executive dashboard

## 4. Key findings

**Which products earn the most revenue?** P203 (Fashion, about N86K), P085 and P229 (both Food) and P006 lead. By category, Fashion is the largest at 38% of revenue, then Food (33%) and Electronics (28%).

**Which customers contribute the most?** C0223, C0773 and C0599 are the top three. Revenue is widely spread: the top 10 customers hold only about 5% of revenue, and the top 10% of customers hold about 27%, so there is little dependence on a few accounts.

**Which warehouses operate most efficiently?** W01 (Abuja), W10 (Kano) and W07 (Lagos) have the lowest delay rates, at about 59% to 60%. W05, W09 and W03 are weakest at 67% to 68%. The spread is narrow, so no warehouse is performing well in absolute terms. Workload also varies: W14 and W15 handle about 35 to 38 shipments per staff member, against about 20 at W13 and W03.

**Which suppliers perform well?** The data has no supplier link, so we used a documented assumption (see Section 6). On that basis Supplier 116, 65 and 85 score highest, and Supplier 108, 73 and 109 lowest. These scores are illustrative only.

**Which products have the highest return rates?** P032 (21%), P017 (19%) and P108 (18%) stand out. Category return rates are similar at 9.4% to 9.6%. The reasons are Late Delivery (185), Wrong Item (163) and Damaged (152).

**Which regions have the most delays?** Kano is worst (65.7% delayed, 3.64 days), then Abuja and Ibadan (about 64%) and Lagos (61%). The gap between regions is small.

**Which months are busiest?** August is the peak, with 654 orders and the highest revenue (about N997K). April and June are also strong. September is a partial month and should not be read as a drop.

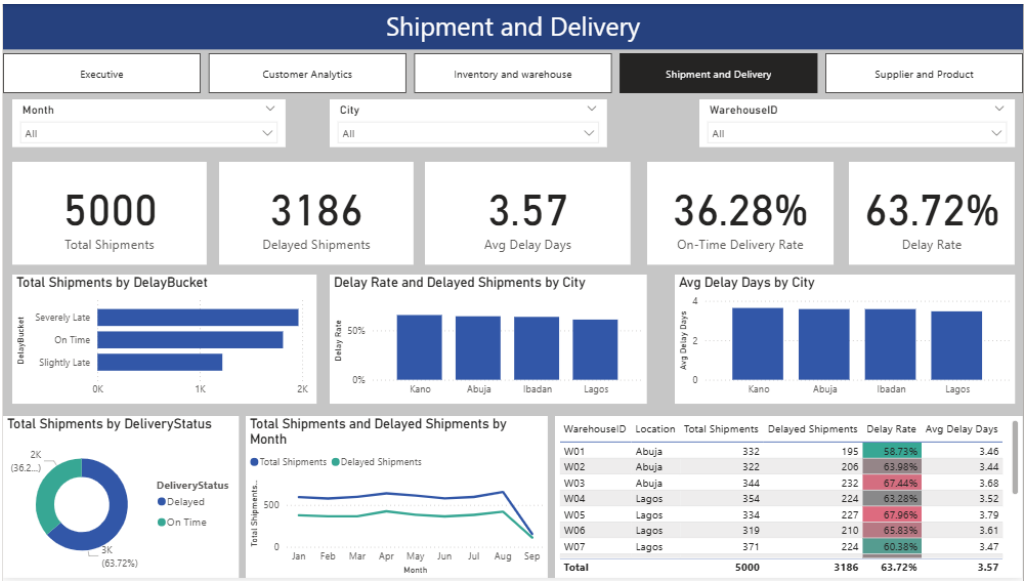



Figure 2: Shipment and delivery dashboard

## 5. KPIs needing attention and inefficiencies

- **On-time delivery (36%) and average delay (3.6 days):** the largest gap in the business, and every region is affected.
- **Inventory imbalance:** about 1M units in stock and a turnover of 0.03, yet 264 warehouse and product pairs are below 100 units. Stock is in the wrong places.
- **Order completion:** roughly a third of orders are delivered, a third pending and a third cancelled, which limits revenue.
- **Returns driven by service:** late delivery is the top return reason, linking delays to lost revenue.
- **Coverage gap:** Ibadan has the most customers (262) but no warehouse. Lagos has 8 warehouses for 246 customers.

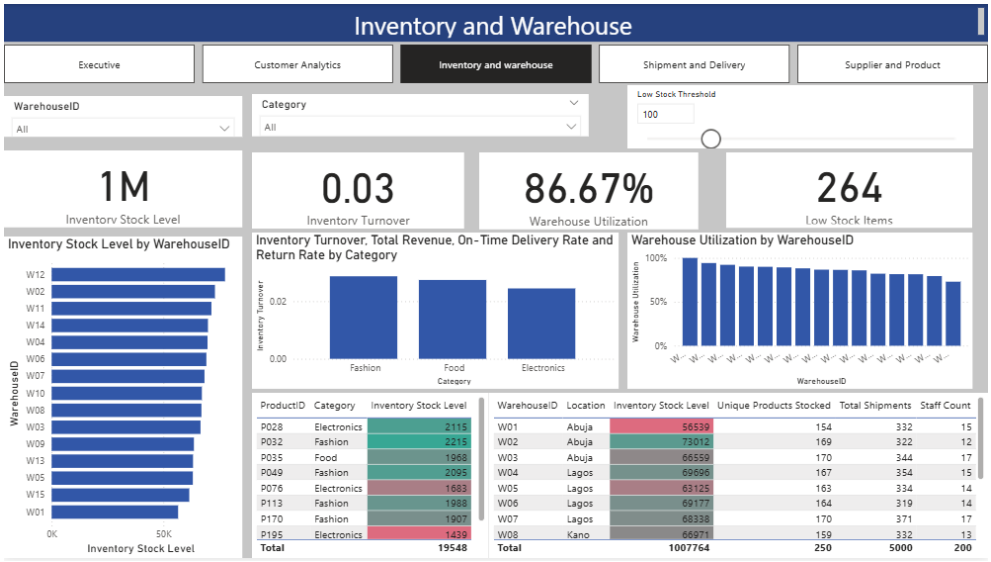


Figure 3: Inventory and warehouse dashboard

## 6. Business opportunities and recommendations

| # | Recommendation | Dashboard evidence |
|---|---|---|
| 1 | Set a delivery recovery plan with a target of 70% on time within two quarters | Only 36% on time; 3.6 day average delay |
| 2 | Introduce delay alerts at 2 days and escalate shipments reaching 3 days | Delays spread evenly across 0 to 7 days |
| 3 | Copy practices from W01, W10 and W07 into W05, W09 and W03 | Delay rates 59% vs 68% |
| 4 | Rebalance stock between warehouses and restock the 264 low-stock pairs | Low Stock Items card; 0.03 turnover |
| 5 | Reduce slow-moving stock by tying purchase orders to sales | Turnover 0.03 on about 1M units |
| 6 | Rebalance staffing at W14, W15 and W12 toward W13 and W03 levels | Shipments per staff about 35 vs 20 |
| 7 | Open or partner for an Ibadan hub to shorten routes | Ibadan has most customers, no warehouse |
| 8 | Add carrier performance reviews and penalties for late delivery | Late delivery is the top return reason |
| 9 | Run a quality check on P032, P017 and P108 and tighten packing and picking | Return rates 18% to 21% vs 9.5% average |
| 10 | Launch a loyalty offer for one-time buyers to lift retention above 48% | Retention 48.3% |
| 11 | Grow Fashion and Food bundles and plan stock ahead of August peaks | Fashion 38% of revenue; August peak |
| 12 | Fix the order pipeline: investigate cancellations and pending backlog | Only about one third of orders delivered |
| 13 | Track the five dashboard KPIs weekly and review monthly with leadership | On-time rate, delay, turnover, returns, retention |

## 7. Assumptions and data limitations

- **Suppliers:** the data has no link between suppliers and products or shipments. Suppliers were treated as a reference table, and for the supplier view each product was assigned to one of 150 suppliers in a fixed, repeating pattern. Supplier scores are therefore illustrative and not real performance. The score combines on-time rate (70%) and return rate (30%).
- **Warehouse capacity:** not provided. Utilisation is an estimate: a warehouse's stock compared with the highest-stocked warehouse (average 86.7%). Unique products and shipments handled are shown alongside it.
- **Inventory:** repeated warehouse and product rows were treated as stock movements, not duplicates, and summed.
- **Delayed:** defined as 3 or more delay days; on time as 0 to 2. Average delivery time uses delay days because no ship or delivery dates exist.
- **Revenue:** quantity multiplied by unit price, delivered orders only. 236 orders have no items and add no revenue.

- **Returns:** 335 of 500 returns are on cancelled or pending orders, which is inconsistent. Return rate is orders with a return divided by all orders.
- **Data dictionary:** lists file names only, so meanings were inferred from column names. Data covers January to September 2026, with September partial.
- **Improvements:** record ship and delivery dates, warehouse capacity, supplier IDs on products and shipments, and return validation.

## 8. Conclusion

SwiftLog's growth is not the problem; its delivery process is. Fixing on-time delivery would cut returns, protect revenue and raise customer retention. Inventory and staffing can then be rebalanced using the dashboard. The five-page Power BI report gives leadership a repeatable way to monitor these measures.